

Slide 1: Indexing Concepts

23IT201 • Database Management Systems • Module 9: Indexing & B+ Trees

Indexes speed up data retrieval by maintaining auxiliary search structures at the cost of additional disk space and update overhead.

Slide 2: B+ Tree Data Structure

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Balanced search tree where internal nodes contain search keys and leaf nodes contain actual data pointers with sequential leaf linking.

Slide 3: Why B+ Trees for Disk Storage

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High node fan-out minimizes tree height (typically 3-4 levels for millions of rows), reducing disk I/O operations.